

EMPLOYEE TRAINING EFFECTIVENESS

A Power BI Dashboard Project — Build Guide & Insights

Dataset: 5,000 employee training records • 10 columns • CSV & XLSX

Why This Dashboard Exists

1

Measure training ROI

Track how much employee skill scores improve after training, relative to program cost.

2

Spot completion gaps

Identify which departments, regions, or programs have low completion or high drop-off.

3

Guide L&D investment

Show leadership which programs deliver the strongest learning gains and satisfaction.

4

Demonstrate BI skills

Show end-to-end capability: data modeling, DAX, calculated columns, and storytelling.

Business question: “Are our training programs actually improving employee performance, and is the spend justified?”

Dataset Overview — 10 Columns, 5,000 Rows

Column	Type	Description
EmployeeID	Text	Unique employee identifier (EMP10001–EMP15000)
Department	Text	Sales, IT, HR, Finance, Operations, Marketing, Support, R&D
Region	Text	North, South, East, West, Central
TrainingProgram	Text	One of 10 training courses offered
EnrollmentDate	Date	Date employee enrolled (2024–2025)
CompletionStatus	Text	Completed / In Progress / Dropped
PreTrainingScore	Number	Assessment score before training (0–100)
PostTrainingScore	Number	Score after training (blank if not completed)
TrainingCost_INR	Number	Cost of training per employee (₹)
SatisfactionRating	Number	Post-training feedback score (1–5)

QUICK FACTS

5,000

Total records

10

Data columns

8

Departments

5

Regions

10

Training programs

24 mo

Date range (2024–25)

6 Steps to Build the Dashboard in Power BI

1 Get & Clean Data

Import the CSV/XLSX via Get Data. Fix data types (dates as Date, scores/cost as Decimal). Handle blanks in PostTrainingScore in Power Query.

2 Model the Data

Keep a single flat table, or split into Fact_Training + Dim tables. Build a Date table and mark it as the official Date table.

3 Add Calculated Columns

Create row-level fields for filtering/grouping: Score Improvement, Enrollment Month, Outcome Flag, Cost Bucket.

4 Write DAX Measures

Build reusable measures: Completion Rate %, Avg Score Improvement, Avg Satisfaction, Total Cost, Cost per Completion.

5 Design the Report Page

Add KPI cards, bar/column charts, a matrix, and slicers. Apply a consistent theme and clear titles.

6 Test, Format & Publish

Check slicer interactions, format numbers/currency, add tooltips, test mobile layout, then publish.

Calculated Columns

Row-level columns added in Power BI to enable grouping, filtering, and conditional formatting.

Score Improvement

Drives the improvement histogram & top-programs ranking.

```
Score Improvement =  
IF(ISBLANK('Training'[PostTrainingScore]), BLANK(),  
    'Training'[PostTrainingScore] - 'Training'[PreTrainingScore])
```

Enrollment Month

Powers the monthly enrollment trend line & slicer.

```
Enrollment Month =  
FORMAT('Training'[EnrollmentDate], "MMM YYYY")
```

Outcome Flag

Simplifies status into a 3-tier label for KPI icons/colors.

```
Outcome Flag =  
SWITCH('Training'[CompletionStatus],  
    "Completed", "On Track", "Dropped", "At Risk", "In Progress")
```

Cost Bucket

Groups spend into bands for the cost-distribution chart.

```
Cost Bucket =  
IF('Training'[TrainingCost_INR]<200, "Low",  
    IF('Training'[TrainingCost_INR]<400, "Medium", "High"))
```

Core DAX Measures

Measures aggregate across whatever filters/slicers are applied — used directly on KPI cards and charts.

Completion Rate %

```
Completion Rate % =  
DIVIDE(  
    CALCULATE(COUNTROWS('Training'),  
        'Training'[CompletionStatus]="Completed"),  
    COUNTROWS('Training'))
```

Avg Score Improvement

```
Avg Score Improvement =  
AVERAGE('Training'[Score Improvement])
```

Avg Satisfaction

```
Avg Satisfaction =  
AVERAGE('Training'[SatisfactionRating])
```

Total Training Cost

```
Total Training Cost =  
SUM('Training'[TrainingCost_INR])
```

Cost per Completion

```
Cost per Completion =  
DIVIDE([Total Training Cost],  
    CALCULATE(COUNTROWS('Training'),  
        'Training'[CompletionStatus]="Completed"))
```

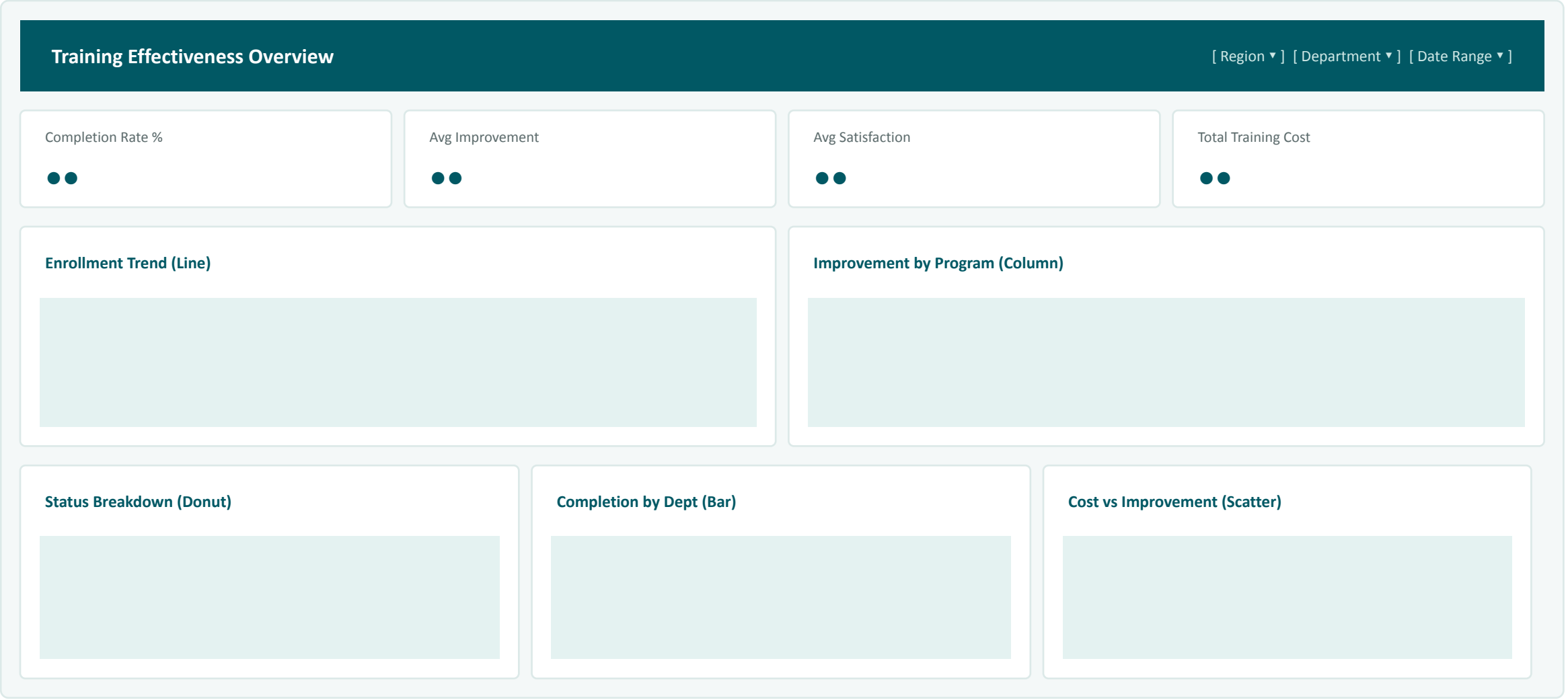
Drop-off Rate %

```
Drop-off Rate % =  
DIVIDE(  
    CALCULATE(COUNTROWS('Training'),  
        'Training'[CompletionStatus]="Dropped"),  
    COUNTROWS('Training'))
```

Chart & Visual Specifications

Visual	Type	Fields Used	Purpose
KPI Cards	Card	Completion Rate %, Avg Improvement, Avg Satisfaction, Total Cost	At-a-glance headline metrics
Enrollment Trend	Line chart	Enrollment Month (X) vs Count of EmployeeID (Y)	Track enrollment volume over time
Completion by Dept	Clustered bar	Department (Y) vs Completion Rate % (X)	Compare departments' completion
Improvement by Program	Column chart	TrainingProgram (X) vs Avg Score Improvement (Y)	Rank programs by learning gain
Cost vs Improvement	Scatter chart	TrainingCost_INR (X), Score Improvement (Y), Program (legend)	Spot cost-effective programs
Status Breakdown	Donut chart	CompletionStatus, Count of EmployeeID	Show Completed/In Progress/Dropped mix
Detail Matrix	Matrix/Table	Department, Region, Program, all key measures	Drill-down for analysts
Filters	Slicers	Region, Department, EnrollmentDate (range)	Let users filter the whole page

Suggested Page Layout



Data Findings & Insights

Based on the 5,000-row generated dataset (illustrates the kind of storytelling this dashboard supports).

77.1%

Overall completion rate

+21.7 pts

Avg score improvement (completed)

4.07 / 5

Avg satisfaction (completed)

₹15.4L

Total training spend

1

Leadership Essentials and Cloud Computing Basics produced the highest average score gains (~22 pts), while Communication Skills trailed slightly (~21.2 pts) — gains are fairly even across programs.

2

R&D and Finance have the highest completion rates (~79%), while HR is lowest (~74%) — a possible signal to review HR scheduling or workload conflicts.

3

Employees who dropped out rated satisfaction far lower on average (2.65/5) than those who completed (4.07/5), reinforcing that early engagement drives both completion and sentiment.

4

Customer Service Excellence, Leadership Essentials, and Python for Data Analysis are the top 3 programs by enrollment (500+ each), indicating strong demand to plan capacity around.

SUMMARY & TAKEAWAYS

What This Project Demonstrates

- End-to-end workflow: raw data → Power Query cleanup → data model → DAX → visual storytelling.
- Hands-on use of calculated columns vs. measures, and when to use each.
- Business framing: every chart ties back to a real L&D decision (completion, cost, satisfaction).
- Result: training completion is strong (77%) and effective (+21.7 avg pts), but HR completion and early drop-off satisfaction are clear areas to act on.

Resume line: “Built an end-to-end Power BI dashboard analyzing training effectiveness across 5,000 employee records — designed calculated columns and DAX measures to track completion rate, score improvement, and training ROI.”